

CSCD608: Advanced Computer Vision — Technical Documentation & User Manual

Complete A-to-Z Technical Reference, Geometry Formulations, and API Guide

Course: CSCD608: Advanced Computer Vision (3 Credits)	Examination: 2025/2026 Second Semester
Question: Question 1 — Feature Matching & Panoramic Stitching	Author: Postgraduate Candidate (MPhil / MSc Computer Science)

Welcome to the comprehensive technical documentation for the **Feature-Based Image Matching and Automatic Panorama Construction System** developed for the CSCD608 postgraduate examination.

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Part I: Getting Started

1. Project Mandate & Scope

- **Degree:** MPhil / MSc Computer Science
- **Course:** CSCD608: Advanced Computer Vision (3 Credits)
- **Semester:** Second Semester Examinations 2025/2026
- **Question:** Question 1 — Feature-Based Image Matching and Automatic Panorama Construction

The goal of this system is to identify corresponding visual regions across two or more overlapping planar or perspective images and automatically composite them into a seamless panorama. The pipeline is engineered entirely using classical computer vision primitives without black-box stitching functions (`cv2.Stitcher_create`) or deep learning models.

2. Examination Requirements Traceability Matrix

Task ID	Examination Specification	Implementation Module	Automated Test / Artifact	Verification Status
REQ-01	Multi-image scene acquisition (≥ 3 views)	src/preprocessing.py	data/raw/	Verified
REQ-02	Aspect-preserving preprocessing & validation	src/preprocessing.py	outputs/baseline/preprocessed_*.png	Verified
REQ-03	Keypoint detection (SIFT & ORB)	src/features.py	tests/test_features.py	Verified
REQ-04	Feature description (Float32 & Binary)	src/features.py	tests/test_features.py	Verified
REQ-05	Descriptor matching with appropriate norms	src/matching.py	tests/test_matching.py	Verified
REQ-06	Initial correspondences visualization	src/visualization.py	outputs/*/raw_matches_*.png	Verified
REQ-07	RANSAC outlier rejection (4-pt DLT consensus)	src/homography.py	tests/test_homography.py	Verified
REQ-08	3×3 Homography matrix estimation & storage	src/homography.py	outputs/*/homographies/	Verified
REQ-09	Perspective image warping & coordinate shift	src/warping.py	tests/test_warping.py	Verified
REQ-10	Multi-image panorama stitching & blending	src/stitching.py	outputs/*/panorama/	Verified
REQ-11	Compare matches Before vs. After RANSAC	src/visualization.py	outputs/*/before_after_*.png	Verified
REQ-12A	In-plane rotation stress test (0° – 180°)	experiments/run_rotation.py	results/rotation_results.csv	Verified
REQ-12B	Multi-scale stress test ($0.5\times$ – $2.0\times$)	experiments/run_scale.py	results/scale_results.csv	Verified
REQ-12C	Perspective viewpoint shear stress test	experiments/run_viewpoint.py	results/viewpoint_results.csv	Verified
REQ-12D	Photometric illumination stress	experiments/run_illumination.py	results/illumination_results.csv	Verified

Task ID	Examination Specification	Implementation Module	Automated Test / Artifact	Verification Status
	test			
REQ-13	Stage-wise execution timing profiling	src/evaluation.py	results/tables/all_results.csv	Verified
REQ-14	Quantitative evaluation of SIFT vs. ORB	scripts/generate_report_tables.py	results/tables/comparison_table.csv	Verified
REQ-15	Complete demonstrable CLI, App & Colab pipeline	scripts/run_pipeline.py , app/server.py	tests/test_pipeline.py	Verified

3. Installation & Quickstart Guide

```
# 1. Clone repository
git clone https://github.com/mhiskall282/Feature-Based-Image-Matching-and-Automatic-Panorama-Construction-Problem.git
cd Feature-Based-Image-Matching-and-Automatic-Panorama-Construction-Problem

# 2. Create and activate virtual environment
python -m venv venv
# Windows: .\venv\Scripts\activate
# Linux/macOS: source venv/bin/activate

# 3. Install requirements
pip install -r requirements.txt

# 4. Launch interactive dashboard
python run_app.py
```

Part II: Computer Vision Mathematics & Pipeline

4. Projective Geometry & 8-DOF Homography

A homography $H \in \mathbb{R}^{3 \times 3}$ maps points between two projective image planes:

$$\begin{matrix} x' & & h_{11} & h_{12} & h_{13} & x \\ y' & \sim & h_{21} & h_{22} & h_{23} & y \\ 1 & & h_{31} & h_{32} & h_{33} & 1 \end{matrix}$$

In inhomogeneous Cartesian coordinates:

$$x' = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}, \quad y' = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$$

Because H is defined up to scale, it has **8 degrees of freedom (DOF)**. The Direct Linear Transformation (DLT) sets up the algebraic constraint $A\mathbf{h} = \mathbf{0}$, which is solved via SVD over consensus inliers.

5. Image Preprocessing & Normalization Protocol

- Aspect-preserving resizing (maximum dimension constrained to 1280 px).
- Grayscale conversion via ITU-R BT.601 standard ($Y = 0.299R + 0.587G + 0.114B$).

- Image buffer integrity validation.

6. Feature Detection: SIFT (DoG) vs. ORB (FAST)

- **SIFT**: Difference-of-Gaussians (DoG) extrema detection across 3 octaves with 3 scale layers per octave. Sub-pixel quadratic Taylor expansion refinement eliminates low-contrast points and unstable edge responses.
- **ORB**: FAST-9 corner detector across an 8-level image pyramid. Harris corner scores rank and retain the top $N = 1000$ points.

7. Feature Description: Gradient Histograms vs. Steered rBRIEF

- **SIFT (128-D Float32)**: 8-bin gradient orientation histograms computed over a 4×4 grid of spatial subregions around the keypoint, normalized to unit L_2 length.
- **ORB (256-Bit Binary)**: Orientation angle $\theta = \text{atan2}(m_{01}, m_{10})$ computed from image moments m_{pq} . Evaluates 256 steered pairwise binary intensity tests $\tau(p; \mathbf{x}_i, \mathbf{y}_i)$.

8. Descriptor Matching & Lowe's Ratio Test

- **SIFT**: Euclidean (L_2) distance with Lowe's ratio test ($d_1 < 0.75 \cdot d_2$).
- **ORB**: Hamming distance with bidirectional cross-check matching.

9. Robust Geometric Estimation via RANSAC

- Randomly sample 4 point pairs.
- Solve candidate H_{cand} via DLT.
- Classify points as inliers if $e_i < 5.0$ px.
- Re-estimate H on all inliers via SVD least-squares refinement.

10. Homography Matrix Diagnostics & Degeneracy Checks

- **Determinant Check**: $\det(H) \in [10^{-6}, 10^6]$.
- **Condition Number**: $\text{cond}(H) < 10^8$.
- **Corner Bounds Sanity**: Verifies warped corners remain within canvas boundaries.

11. Dynamic Canvas Geometry & Perspective Warping

$$x_{offset} = \max(0, -\min(x_{warped})), \quad y_{offset} = \max(0, -\min(y_{warped}))$$

$$T = \begin{bmatrix} 1 & 0 & x_{offset} \\ 0 & 1 & y_{offset} \\ 0 & 0 & 1 \end{bmatrix}, \quad H_{adjusted} = T \cdot H$$

12. Multi-Image Stitching & Distance-Weighted Alpha Blending

In overlap regions, pixel values are blended using Euclidean distance transforms:

$$I_{blend}(x, y) = \frac{D_1(x, y)}{D_1(x, y) + D_2(x, y)} I_1(x, y) + \frac{D_2(x, y)}{D_1(x, y) + D_2(x, y)} I_2(x, y)$$

Part III: Empirical Research & Benchmarks

13. Quantitative 8-Metric Evaluation Framework

- **M1: Keypoint Count**: Total detected interest points per view.

- **M2: Match Count:** Initial filtered correspondences.
- **M3: RANSAC Inlier Count:** Number of geometrically verified correspondences.
- **M4: Inlier Ratio:** Ratio of inliers to good matches ($N_{\text{inliers}}/N_{\text{matches}}$).
- **M5: Execution Timing:** Stage-wise profiling (detection, matching, RANSAC, total).
- **M6: Panorama Quality:** Non-black pixel coverage ratio and Laplacian gradient sharpness.
- **M7: Mean Reprojection Error:** Average pixel distance of inlier projections.
- **M8: Homography Success:** Boolean status with failure root cause logging.

14. 5 Research Benchmark Experiments & Comparative Analysis

Benchmark Experiment	SIFT Inlier Ratio	ORB Inlier Ratio	SIFT Reproj Error	ORB Reproj Error	SIFT Latency	ORB Latency
Baseline (3 Images)	59.64%	37.74%	0.14 px	0.93 px	1.82 s	2.06 s
Rotation (0° – 180°)	94.57%	91.74%	0.17 px	1.15 px	2.24 s	1.87 s
Scale ($0.5\times$ – $2.0\times$)	96.24%	97.23%	0.18 px	0.23 px	2.52 s	2.40 s
Viewpoint Shear	90.55%	84.87%	0.28 px	0.97 px	4.90 s	4.46 s
Illumination ($\Delta\beta, \alpha$)	91.50%	87.43%	0.27 px	0.38 px	4.58 s	4.36 s

15. Before vs. After RANSAC In-Depth Analysis

Before RANSAC, raw matches contain false correspondences. RANSAC eliminates outliers and ensures only valid geometric consensus pairs determine the projective warp.

Part IV: Diagnostics & Recovery

16. Failure Diagnostics Taxonomy (F1–F5)

- **F1: Low Keypoints** ($N < 4$): Textureless scenes or blur. Aborts estimation.
- **F2: Low Overlap** ($N_{\text{matches}} < 4$): Overlap $< 15\%$. Prevents rank-deficient DLT solver.
- **F3: Outlier Dominance** ($> 95\%$): RANSAC fails to find consensus. Flags failure in log.
- **F4: Degenerate Homography:** Collinear points ($\det(H) \notin [10^{-6}, 10^6]$).
- **F5: Depth Parallax:** 3D camera translation in non-planar scenes causes local ghosting.

Part V: Application Interfaces & User Manual

17. Interactive Web Dashboard Manual

- **Preset Scenes:** 1-click sample scene loading.
- **Custom Upload:** Drag-and-drop 2 to 6 photos.
- **Dual Benchmark Mode:** Runs SIFT and ORB side-by-side.
- **Live Stress Playground:** Interactive sliders for rotation, scale, shear, and illumination.

18. CLI Utilities & Automated Benchmark Runners

```
# Run SIFT CLI pipeline
python scripts/run_pipeline.py --algorithm sift --data data/raw --output outputs/sift

# Run all 5 research experiments
python experiments/run_all.py
```

19. Google Colab Integration Guide

Open `notebooks/analysis.ipynb` in Google Colab to run all pipeline stages interactively.

Part VI: Production Cloud Deployment & Verification

20. Cloud Deployment Guide (Render, Vercel, Docker)

- **Render:** Detects `render.yaml` and deploys automatically.
- **Vercel:** Routes through serverless endpoint `api/index.py`.
- **Docker:** `docker build -t panorama-app . && docker run -p 5000:5000 panorama-app`.

21. Pytest Test Suite & Validation Report

```
pytest tests/ -v
# 31 passed in 1.40s (100% passing)
```

22. Academic References

- **Lowe, D. G. (2004).** Distinctive image features from scale-invariant keypoints. **IJCV**, 60(2), 91–110.
- **Rublee, E., et al. (2011).** ORB: An efficient alternative to SIFT or SURF. In **IEEE ICCV** (pp. 2564–2571).
- **Fischler, M. A., & Bolles, R. C. (1981).** Random sample consensus: a paradigm for model fitting. **CACM**, 24(6), 381–395.
- **Brown, M., & Lowe, D. G. (2007).** Automatic panoramic image stitching using invariant features. **IJCV**, 74(1), 59–73.
- **Hartley, R., & Zisserman, A. (2004).** **Multiple View Geometry in Computer Vision** (2nd ed.). Cambridge University Press.